

Amir Mirzai Golpayegani

📍 Calgary, AB, Canada

✉ amirgolpaa24@gmail.com | ✉ amir.mirzaigolpayega@ucalgary.ca | 📞 +1 (587) 439-0436

🌐 linkedin.com/in/amir-golpayegani-218148165 | 🐙 github.com/amirgolpaa24 | 🌐 Portfolio

Summary

M.Sc. Computer Science graduate and junior software developer with experience building Python backends, REST APIs, data pipelines, and user-facing automation tools. Strong foundation in Django, JavaScript, SQL, Git, Docker, testing-minded development, and collaborative problem solving, with hands-on exposure to AI-enabled workflows and scalable data systems.

Education

M.Sc. in Computer Science University of Calgary	GPA: 3.94/4.0	Sep 2022 – Feb 2026 Calgary, Canada
B.Sc. in Computer Engineering Sharif University of Technology	GPA: 3.69/4.0	Sep 2016 – Feb 2021 Tehran, Iran

Technical Skills

Programming: Python, JavaScript, C++, SQL

Backend / Full-Stack: Django, Node.js, REST APIs, PostgreSQL, authentication flows, payment-service logic

Engineering Practices: Git, GitHub, Docker, Linux, GitHub Actions, debugging, documentation, code review readiness

AI / Data: PyTorch, Scikit-learn, TensorFlow, CNNs, LLM workflows, AI agents, NumPy, Pandas, SciPy

Visualization / Automation: Streamlit, Polyscope, OpenGL, GLSL, n8n, OpenClaw, SearXNG, Telegram API

Work Experience

Back-End Developer Asr Gooyesh Pardaz	Jun 2020 – Jul 2022 Tehran, Iran
---	-------------------------------------

- Developed Django REST APIs for NLP, text-to-speech, and speech-to-text service plans backed by PostgreSQL.
- Built authentication features including registration, Google login, password reset, and email/phone verification.
- Implemented backend logic for payment-service integration and storing plan/payment records in a relational database.
- Contributed to a WebRTC-based chatbot system using GPT-style responses and FAQ data for user-facing support flows.

Research Software Developer BigGeo / Mitacs, University of Calgary	Sep 2022 – Feb 2026 Calgary, Canada
--	--

- Built C++/Python software for scalable processing, storage, and retrieval of large scientific datasets.
- Automated SAR and optical imagery preprocessing using GDAL, Rasterio, SNAP, and satellite data APIs.
- Developed structured encoding pipelines that converted raw imagery into queryable tensor files.
- Improved encoding performance with C++ multithreading, reducing 10-tensor processing from 233s to 26s.
- Documented experimental workflows and evaluated storage/retrieval behavior across scaled datasets.

Deep Learning Teaching Assistant Neuromatch Academy	Jul 2026 Remote, United States
---	-----------------------------------

- Mentored an international pod of ~15 students through Neuromatch Academy's intensive three-week Deep Learning course.
- Facilitated hands-on PyTorch tutorials spanning MLPs, CNNs, attention/transformers, diffusion generative models, and reinforcement learning.
- Guided project groups through end-to-end deep learning research projects, from question formulation and literature review to model implementation and final presentations.

Teaching Assistant Department of Computer Science, University of Calgary	Dec 2022 – Apr 2025 Calgary, Canada
--	--

- **Developing Big Data Applications:** Mentored +25 students on AWS-based data pipelines, cloud storage, debugging, and scalable data workflows.
- **Working with Data at Scale:** Supported +20 students with large-scale data processing, data pipelines, and high-volume data analysis.
- **Database Management Systems:** Assisted +75 students with SQL, database design, course projects, grading, and exam support.

Research and Technical Projects

Agentic Job Search Automation Platform OpenClaw, n8n, LLMs, AI Agents, Docker, Workflow Automation, SearXNG, Telegram API	May 2026 – Present
---	--------------------

- Building an agentic AI system to search jobs, analyze job fit, and generate tailored application materials.
- Designed LLM-based workflows for resume/job matching, skill-gap analysis, and application decision support.
- Integrated Google Sheets, Google Drive, and Telegram API to track jobs and send automated application alerts.

• Prototyped the initial workflow in n8n before migrating toward a more flexible OpenClaw-based agent system.	
Persian Text Normalizer for NLP <i>Python, NLP, Text Processing</i>	Oct 2016 – Feb 2021
• Developed a Python-based text normalization toolkit to standardize Persian text for downstream NLP pipelines. • Implemented preprocessing rules for cleaning, formatting, and preparing Persian text data for language-processing tasks.	
Spatiotemporal Earth Observation Data System Using DGGs <i>First-author publication M.Sc. thesis C++, Python, DGGs, Wavelets, B-splines, Time-Series</i>	Sep 2022 – Feb 2026
• Designed a DGGs-based framework for real-time multiresolution management of satellite imagery. • Developed a tensor-based storage model for direct access to arbitrary regions at multiple resolutions. • Implemented retrieval and evaluation workflows for large spatiotemporal datasets.	
DEM Super-Resolution Framework <i>Python, PyTorch, CNNs, Scientific Imaging, Google Earth Engine API, Polyscope, Data Processing and Visualization</i>	May 2024 – Feb 2026
• Trained deep ResNet models for $4\times$ DEM super-resolution, reconstructing 30m elevation data from 120m inputs. • Built a Streamlit pipeline using Rasterio to parallelize GEE DEM tile downloads and systematically collect and preprocess mountainous terrain data worldwide. • Implemented PyTorch training and evaluation workflows with terrain-aware metrics such as RMSE, slope, TRI, TPI, and wavelet-based errors. • Developed a Polyscope-based 3D visualization tool to render DEM tiles from multiple angles and compare reconstruction quality in 3D.	
Publication	
Amir Mirzai Golpayegani , Mahmudul Hasan, Faramarz F. Samavati. <i>Real-Time Multiresolution Management of Spatiotemporal Earth Observation Data Using DGGs</i> . Remote Sensing, 2025. doi:10.3390/rs17040570	
Certificates	
Supervised Machine Learning: Regression and Classification , DeepLearning.AI and Stanford University, Coursera	Oct 2024
Applied Machine Learning in Python , University of Michigan, Coursera	Jul 2020
Languages	
English: Full Professional Proficiency (TOEFL 115/120) Persian: Native Proficiency	
Achievement	
Teaching Assistant Excellence Laureate Department of Computer Science, University of Calgary — awarded for outstanding teaching support (link)	2024/2025
Ranked 17th Nationwide in Iran University Entrance Exam (Konkour) Top 0.001% among all participants	May 2016